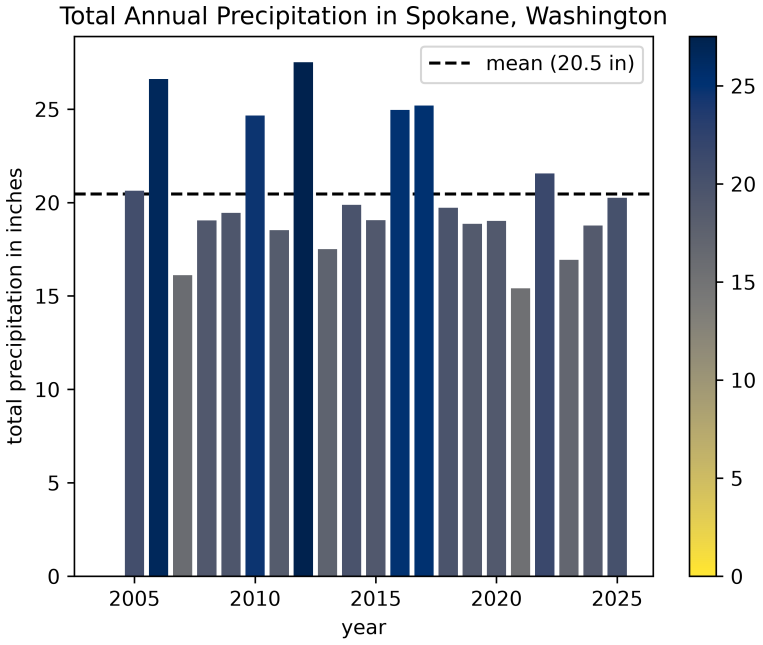
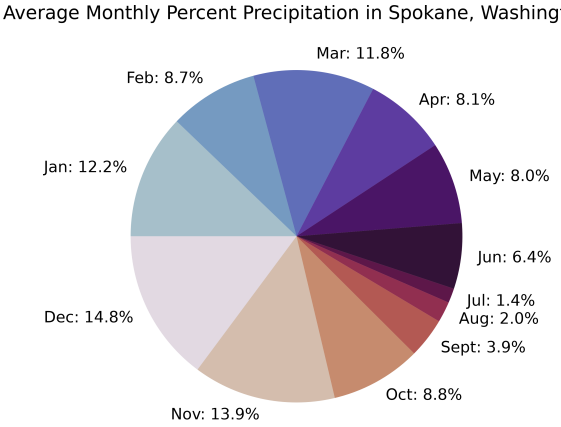
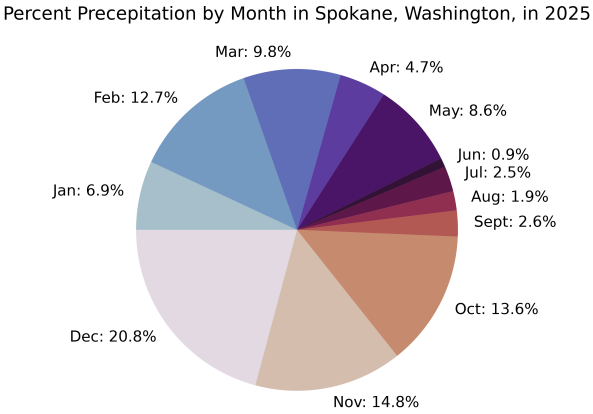


(a)



(b)



(c)

Figure 1: Precipitation trends for Spokane, Washington. (a) shows the total precipitation in inches for each year, with the colors of the bars corresponding to the total precipitation. (b) shows the percent of the total precipitation from 2005 through 2025 each month accounts for. (c) shows the breakdown by percentage of precipitation per month for 2025. All color schemes used, cividis and twilight, are designed to be perceptually uniform and CVD friendly.

Analysis of Precipitation in Spokane, Washington, USA

Summing all the precipitation data for each year, then totaling the precipitation across all years from 2005 through 2025 and dividing by the number of years yielded a mean precipitation for the Spokane area of 20.5 inches. As shown in Figure 1a, 2005, 2006, 2010, 2012, 2016, 2017, and 2022 all had above average precipitation. At around 20 inches, the precipitation in 2025 was barely below the mean. The driest year in the last 20 years was in 2021 with around 15 inches of total precipitation.

Figure 1b displays the percent precipitation per month, calculated by summing the precipitation in each month across all years and dividing by the total precipitation of all years combined. While the percentage

of yearly precipitation per month may vary annually, this graph demonstrates the general trend. According to Figure 1b, in Spokane, Washington, July tends to recieve the least percentage precipitation at 1.4 percent with August at 2.0 percent in a close second. The months with the highest percentage of precipitation are December at 14.8 percent, November at 13.9 percent, January at 12.2 percent, and March at 11.8 percent precipitation. On average, 86.3 percent of Spokane's precipitation occurs between the months of October and May.

The distribution of precipitation per month in 2025 had some significant deviations from the standard. As seen in Figure 1c, the largest variations were in December with a difference of 6 percent, June, with a difference of 5.5 percent, and January, with a difference of 5.3 percent. August was closest to the predicted percentage with a difference of .1 percent. In 2025, June through September only made up 7.9 percent of the annual precipitation. The percentages for Figure 1c were calculated by summing the precipitation for each month of 2025 and dividing by the total precipitation for the 2025 year.